



JURONG JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

CANDIDATE
NAME

CLASS

18S

EXAM INDEX

CHEMISTRY

9729/01

Higher 2

Paper 1 Multiple Choice

13 September 2018

1 hour

Candidates answer on separate paper.

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and exam index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** or **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1 The successive ionisation energies (IE) of two elements, **E** and **J**, are given below:

IE/ kJ mol ⁻¹	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
E	550	1065	4138	5500	6910	8760	10230	11800
J	1140	2103	3470	4560	5760	8550	9940	18600

What is the likely formula of the compound that is formed when **E** reacts with **J**?



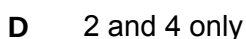
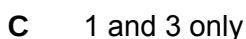
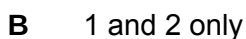
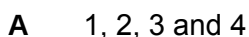
- 2 Carbon, silicon and germanium are Group 14 elements and they all exist in a structure similar to diamond.

The given table shows the bond lengths in these structures.

Element X	C	Si	Ge
Bond length X–X / nm	0.154	0.234	0.244

Which of the following explain why the bond length increases down the group?

- 1 Degree of orbital overlap decreases down the group.
- 2 Atomic radius increases down the group.
- 3 Nuclear charge increases down the group.
- 4 Electronegativity decreases down the group.



- 3 Which of the following **cannot** be explained by hydrogen bonding?



- 4 The percentage by mass of magnesium in a mixture of magnesium chloride and magnesium nitrate was found to be 21.25%.

What is the mass of magnesium chloride present in 100 g of the mixture?

- | | | | |
|----------|------|----------|------|
| A | 47 g | B | 51 g |
| C | 53 g | D | 56 g |

- 5 When 10 cm³ of a gaseous hydrocarbon **X** was burned in 70 cm³ of oxygen, the final gaseous mixture contained 30 cm³ of carbon dioxide and 20 cm³ of unreacted oxygen.

What is the formula of hydrocarbon **X**?

[All gaseous volumes are measured under identical conditions.]

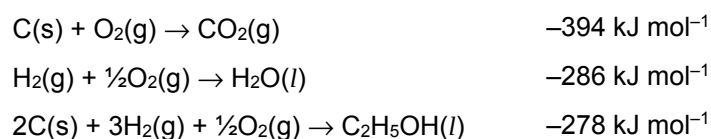
- | | | | |
|----------|-------------------------------|----------|--------------------------------|
| A | C ₂ H ₆ | B | C ₃ H ₆ |
| C | C ₃ H ₈ | D | C ₄ H ₁₀ |

- 6 An ion of metal **L** can be oxidised by potassium manganate(VII) in acid solution to form **LO₃⁻**. In an experiment, 1.25×10^{-3} mol of the ion of **L** required 37.5 cm³ of 0.0200 mol dm⁻³ potassium manganate(VII) for complete reaction.

What is the initial oxidation state of the ion of **L** given that potassium manganate(VII) is reduced to Mn²⁺?

- | | | | |
|----------|----|----------|----|
| A | +1 | B | +2 |
| C | +3 | D | +4 |

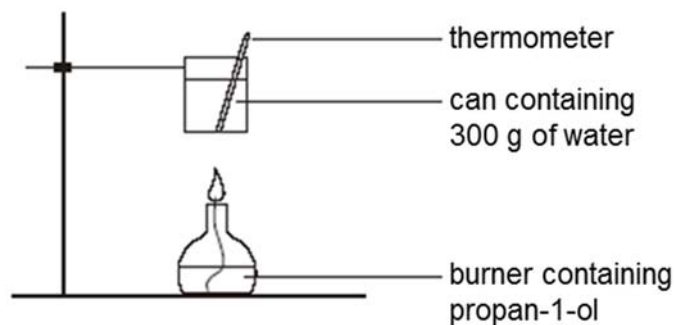
- 7 Given the following standard enthalpy changes,



What is the standard enthalpy change of combustion of liquid ethanol, C₂H₅OH?

- | | | | |
|----------|----------------------------|----------|----------------------------|
| A | -402 kJ mol ⁻¹ | B | -758 kJ mol ⁻¹ |
| C | -1368 kJ mol ⁻¹ | D | -1924 kJ mol ⁻¹ |

- 8 A student used the set-up below to heat a can containing 300 g of water.



The following data were recorded:

$$\begin{aligned}\text{mass of propan-1-ol burnt} &= m \text{ g} \\ \text{change in temperature of water} &= \Delta T \text{ }^{\circ}\text{C}\end{aligned}$$

Given that:

$$\begin{aligned}\text{relative molecular mass of propan-1-ol} &= 60.0 \\ \text{enthalpy change of combustion of propan-1-ol} &= -2021 \text{ kJ mol}^{-1} \\ \text{specific heat capacity of water} &= c \text{ J g}^{-1} \text{ K}^{-1}\end{aligned}$$

What is the efficiency of this heating process?

$$\begin{array}{ll}\text{A} & \frac{m \times 2021 \times 1000}{300 \times c \times \Delta T \times 60.0} \times 100\% \\ \text{B} & \frac{m \times c \times \Delta T \times 60.0}{300 \times 2021 \times 1000} \times 100\% \\ \text{C} & \frac{300 \times c \times \Delta T \times 60.0}{m \times 2021} \times 100\% \\ \text{D} & \frac{300 \times c \times \Delta T \times 60.0}{m \times 2021 \times 1000} \times 100\%\end{array}$$

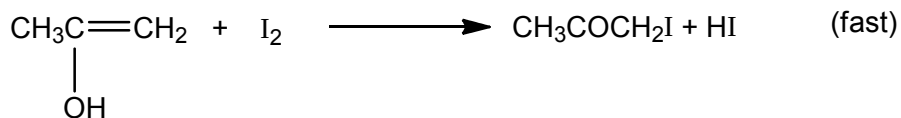
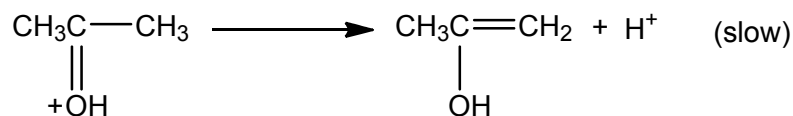
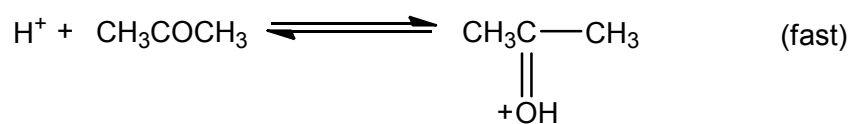
- 9 The experimental results obtained for the reaction between X and Y at constant temperature are given in the table below.

Experiment	[X] / mol dm ⁻³	[Y] / mol dm ⁻³	initial rate / mol dm ⁻³ s ⁻¹
1	0.3	0.2	4.00 x 10 ⁻⁴
2	0.6	0.4	1.60 x 10 ⁻³
3	0.6	1.2	1.44 x 10 ⁻²

What is the rate equation for this reaction?

$$\begin{array}{ll}\text{A} & \text{Rate} = k[\text{Y}]^2 \\ \text{B} & \text{Rate} = k[\text{X}]^2 \\ \text{C} & \text{Rate} = k[\text{X}]^2[\text{Y}] \\ \text{D} & \text{Rate} = k[\text{X}][\text{Y}]^2\end{array}$$

- 10 The mechanism for the iodination of propanone in aqueous acid involves the following steps.



Which of the following statements are true?

- 1 The overall order of the reaction is 1.
- 2 The acid acts as a catalyst.
- 3 The rate equation is $\text{rate} = k[\text{CH}_3\text{COCH}_3][\text{H}^+]$
- 4 The rate of the reaction is not affected by a change in the iodine concentration.

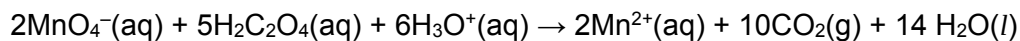
- A** 1, 2 and 4 only
B 2, 3 and 4 only
C 1 and 2 only
D 2 and 3 only

- 11 The rate of removal of *Aspirin*, a pain-killer drug, from the body is a first order reaction with a half-life of 2.0 h.

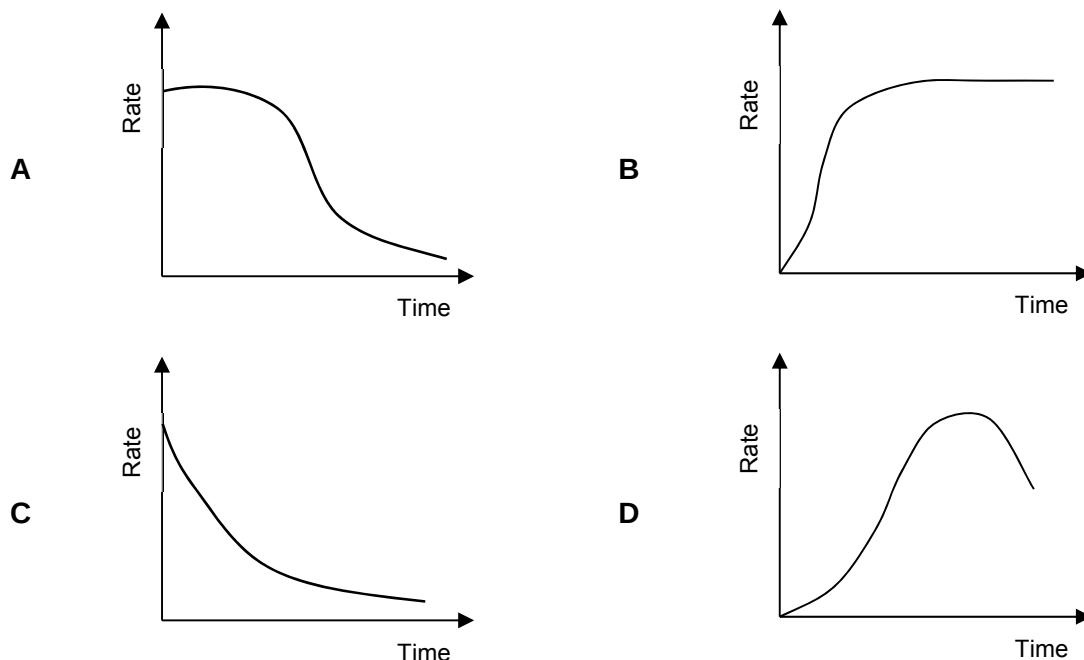
How long does it take for 87.5% of *Aspirin* to be removed from the body?

- A** 6.0 hours **B** 2.0 hours
C 1.0 hours **D** 0.4 hours

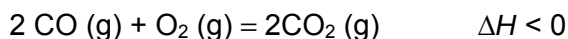
- 12 The reaction between potassium manganate(VII) and ethanedioic acid is an example of auto-catalytic reactions, in which one of the products catalyses the reaction.



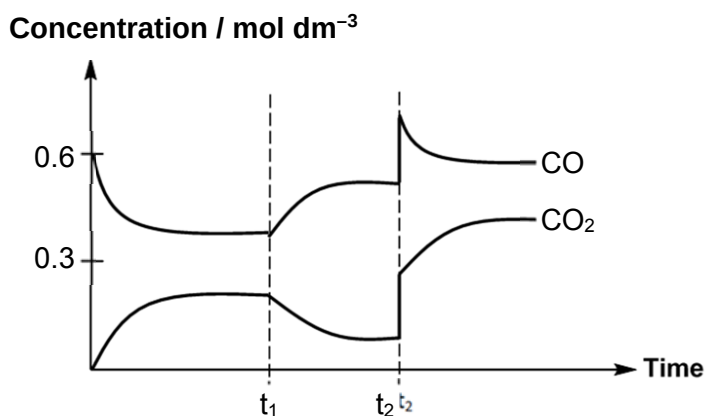
Which graph correctly represents the kinetics of this reaction?



- 13 At a temperature T K, 0.60 mol dm^{-3} of CO and 0.30 mol dm^{-3} of O_2 were introduced into a 5 dm^3 vessel and allowed to reach equilibrium.



The graph below shows the changes in the concentration of CO and CO_2 in the system with time. A change was made to the system at time, t_1 and t_2 .

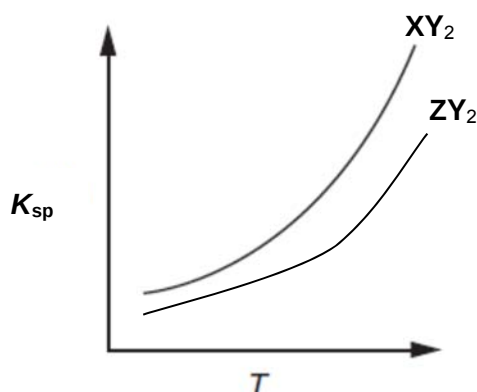


What were the changes made at time, t_1 and t_2 ?

	t_1	t_2
A	A catalyst was added	Volume of the system is increased
B	The temperature was increased	Volume of the system is decreased
C	Some O_2 was removed	An inert gas was added at constant volume
D	The temperature was decreased	More O_2 was added

- 14 XY_2 and ZY_2 are sparingly soluble salts containing Z^{2+} , X^{2+} and Y^- ions.

The solubility product, K_{sp} , of both sparingly soluble salts varies with temperature as shown in the diagram below.



Which conclusions can be drawn from the information?

- 1 The enthalpy change of solution for both salts is endothermic.
- 2 Adding Y^- to the solution of XY_2 decreases the K_{sp} of XY_2 .
- 3 Adding Y^- to the solution of ZY_2 decreases the solubility of ZY_2 .
- 4 Given a solution with $[\text{X}^{2+}] = [\text{Z}^{2+}]$, ZY_2 is precipitated out first on adding Y^- .

- A 1 and 3 only
 B 3 and 4 only
 C 1, 2 and 4 only
 D 1, 3 and 4 only

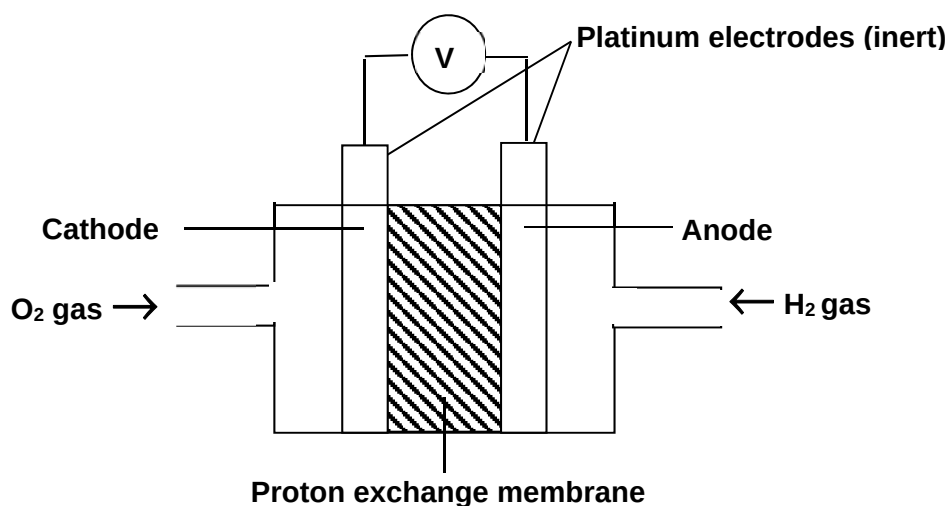
- 15 A metal object was electroplated with chromium using an aqueous electrolyte of chromium(III) chloride and a graphite electrode. A current of 2.0 A was passed through the electroplating cell for 45 minutes.

What may be derived from the information given above?

- 1 The object to be electroplated was connected to the negative terminal of the battery.
- 2 The object increased in mass by 0.97 g.
- 3 Oxygen gas was evolved at the anode.

- A 1, 2 and 3
 B 1 and 2 only
 C 2 and 3 only
 D 1 only

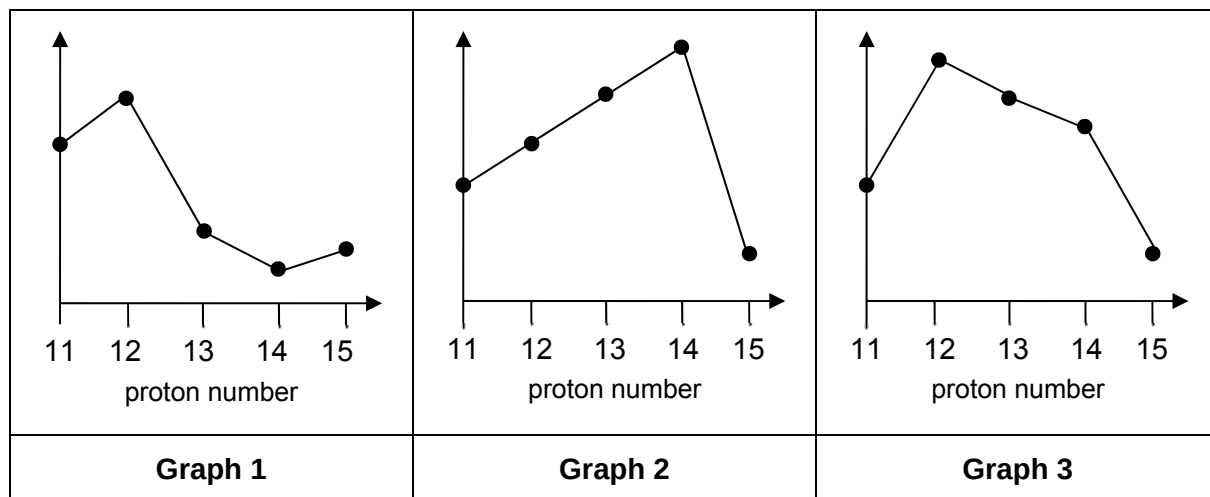
- 16 A hydrogen fuel cell as illustrated below has a typical e.m.f. of 1.23 V.



Which of the following is true regarding the operation of this fuel cell?

- 1 The e.m.f can be increased by increasing the pressure of oxygen gas to 2 atm.
 - 2 The electrode is platinised to increase the rate of reaction, but e.m.f. is not affected.
 - 3 The proton exchange membrane allows the passage of H^+ ions in order to complete the circuit and maintains electrical neutrality.
- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 1 only

- 17 The following graphs show how three properties of the elements, Na to P, and their compounds, vary with proton number.



What properties are shown by the three graphs?

	Graph 1	Graph 2	Graph 3
A	Melting point of oxide	Melting point of chloride	Electrical conductivity of element
B	Melting point of oxide	Melting point of element	Melting point of chloride
C	Melting point of chloride	Electrical conductivity of element	Melting point of oxide
D	Melting point of chloride	Melting point of element	Melting point of oxide

- 18 Which property of Group 2 elements (magnesium to barium) or their compounds increases with increasing proton number?

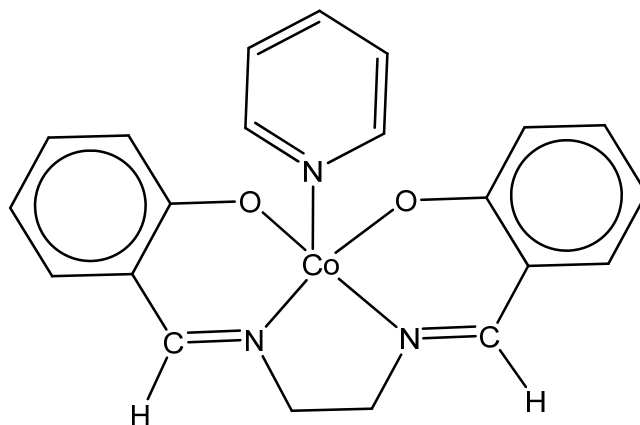
- A the stability of the carbonates to heat
- B the magnitude of the enthalpy change of hydration of the metal ions
- C the acidity of aqueous solutions of the chlorides
- D the melting points of the elements

- 19 For the sequence hydrogen iodide, hydrogen bromide and hydrogen chloride, which of the following properties show a decreasing trend?

- 1 thermal stability
- 2 acidity
- 3 ease of oxidation

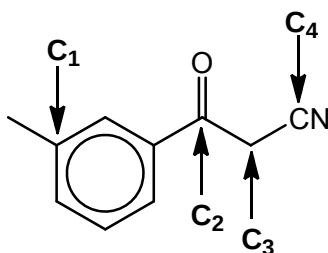
- A 1 only
- B 1 and 2 only
- C 2 and 3 only
- D 1, 2 and 3

- 20 The following cobalt complex is known to be the functional model for biological oxygen carrier.



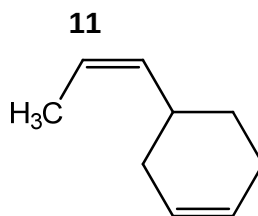
What is the electronic configuration of the cobalt cation in the above complex?

- A [Ar] 3d⁶
 B [Ar] 3d⁷
 C [Ar] 3d⁵ 4s²
 D [Ar] 3d⁷ 4s²
- 21 Transition metals like platinum and rhodium are found in catalytic converters fitted into cars. Which of the following properties best explains the role of transition metals in this use?
- A Transition metals can exhibit variable oxidation states in their compounds as 3d and 4s electrons have similar energies.
 B Transition metals form coloured ions due to absorption of energy in the visible light region to promote an electron from a lower to a higher energy 3d orbital.
 C Transition metals have very high melting points because both 3d and 4s electrons are involved in forming strong metallic bond.
 D Transition metals have partially filled 3d orbitals which are available for adsorption of reactant molecules.
- 22 What is the hybridisation of the various carbon atoms in the following molecule?



- | | C ₁ | C ₂ | C ₃ | C ₄ |
|---|-----------------|-----------------|-----------------|-----------------|
| A | sp ² | sp ³ | sp | sp ² |
| B | sp ² | sp ² | sp ³ | sp |
| C | sp ² | sp ² | sp | sp ³ |
| D | sp ³ | sp ² | sp ³ | sp |

23

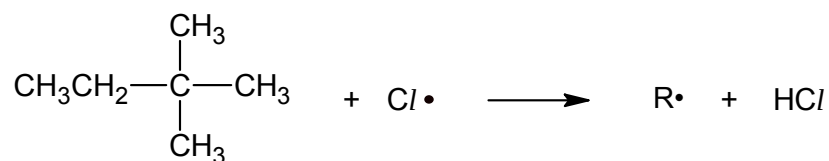


How many stereoisomers does the above molecule have?

A 2**B** 4**C** 6**D** 8

- 24** When heated with chlorine, the hydrocarbon 2,2-dimethylbutane undergoes free radical substitution.

In a propagation step, the free alkyl radical $R^\bullet$ is formed.



How many different forms of $R^\bullet$ are possible?

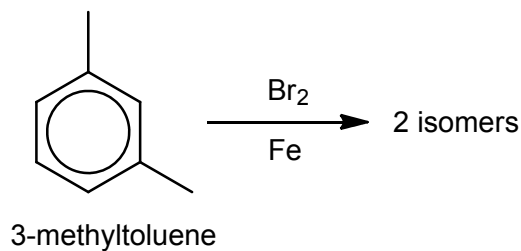
A 1**B** 2**C** 3**D** 4

- 25** Which one of the following statements regarding organic reaction mechanisms is **correct**?

- 1 Electrophilic addition of alkenes involves a carbocation intermediate.
- 2 Electrophilic substitution of benzene involves an intermediate with 4 π electrons.
- 3 Nucleophilic substitution of tertiary halogenoalkanes involves a carbocation intermediate.
- 4 Nucleophilic addition of a carbonyl compound involves an alkoxide intermediate.

A 1, 2, 3 and 4**B** 1 and 2 only**C** 2 and 3 only**D** 3 and 4 only

- 26 When 3-methyltoluene is treated with bromine in the presence of iron filings in the dark, a mixture of two mono-brominated isomers is formed.



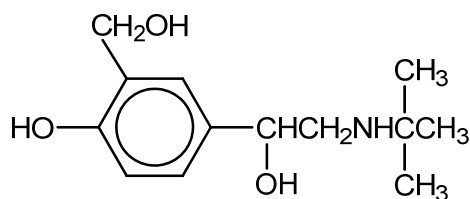
What are the structures of these two isomers?

- A** and
- B** and
- C** and
- D** and

- 27 When ethyl ethanoate undergoes hydrolysis with dilute sulfuric acid in the presence of H_2^{18}O , a mixture of two products is formed. Which of the following pairs correctly gives the structures of the two products?

- A** $\text{CH}_3\text{CO}^{18}\text{OH}$ and $\text{CH}_3\text{CH}_2^{18}\text{OH}$
- B** CH_3COOH and $\text{CH}_3\text{CH}_2^{18}\text{OH}$
- C** $\text{CH}_3\text{C}^{18}\text{OOH}$ and $\text{CH}_3\text{CH}_2\text{OH}$
- D** $\text{CH}_3\text{CO}^{18}\text{OH}$ and $\text{CH}_3\text{CH}_2\text{OH}$

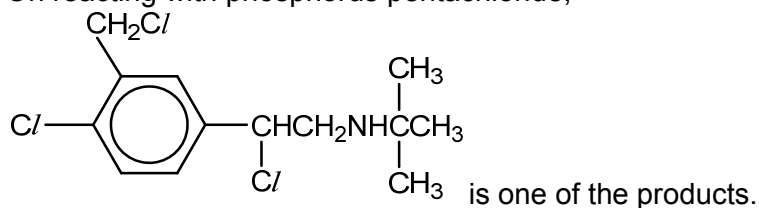
- 28 Albuterol dilates the airways of the lung and is used for treating asthma and other conditions of the lung.



Albuterol

Which of the following statements is true about albuterol?

- 1 On addition of $\text{Na}_2\text{CO}_3(\text{aq})$, effervescence of CO_2 is produced.
- 2 On heating one mole of albuterol with $\text{NaOH}(\text{aq})$, one mole of NaOH is used up.
- 3 On reacting with phosphorus pentachloride,



- A 1 only
 B 2 only
 C 1 and 3 only
 D 2 and 3 only

- 29 A sample of bromoethane was warmed with ethanolic silver nitrate, and a cream precipitate was observed after about 4 minutes.

Under similar reaction conditions, which one of the following compounds will result in precipitate formation only after 8 minutes?

- | | |
|----------------|--------------------|
| A iodoethane | B ethanoyl bromide |
| C bromobenzene | D chloroethane |

- 30 Which of the following gives the correct order of decreasing basic strength?

- A $\text{CH}_3\text{CH}_2\text{NH}_2$, NH_3 , $(\text{CH}_3\text{CH}_2)_2\text{NH}$
 B $\text{CH}_3\text{CH}_2\text{NH}_2$, $(\text{CH}_3\text{CH}_2)_2\text{NH}$, $\text{C}_6\text{H}_5\text{NH}_2$
 C $(\text{CH}_3\text{CH}_2)_2\text{NH}$, NH_3 , $\text{C}_6\text{H}_5\text{NH}_2$
 D $\text{C}_6\text{H}_5\text{NH}_2$, NH_3 , $\text{CH}_3\text{CH}_2\text{NH}_2$

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2018 Paper 1 answer key

1	A	2	B	3	B	4	C	5	C
6	B	7	C	8	D	9	A	10	B
11	A	12	D	13	B	14	D	15	A
16	A	17	D	18	A	19	C	20	B
21	D	22	B	23	B	24	C	25	A
26	C	27	D	28	B	29	D	30	C